```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein (HH)
  Ansatz          : warm: None; ADAPT: phase1_v1; final: matched-family replay
  Drive enabled   : False

  t (hopping)     : 1.0
  U (interaction) : 2.0
  dv (disorder)   : 0.0

  L_values        : 2,3
  omega0          : 1.0
  g_ep            : 1.0
  n_ph_max        : 1
  ordering        : blocked
  boundary        : periodic
  dynamics_pages  : suzuki2,cfqm4 macro-step only

  Command:
    /opt/anaconda3/bin/python pipelines/hardcoded/hh_staged_circuit_report.py

============================================================
```

```
Combined HH staged circuit report

L values: 2, 3
Each L section includes:
- parameter manifest
- stage summary
- warm HH-HVA circuit
- ADAPT circuit
- matched-family replay circuit
- Suzuki2 macro-step circuit
- CFQM4 macro-step circuit

Representative pages preserve PauliEvolutionGate blocks.
Expanded pages decompose one level only for readability.
```

```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein (HH)
  Ansatz          : warm: hh_hva_ptw; ADAPT: phase1_v1; final: matched-family replay
  Drive enabled   : False

  t (hopping)     : 1.0
  U (interaction) : 2.0
  dv (disorder)   : 0.0

  L               : 2
  omega0          : 1.0
  g_ep            : 1.0
  n_ph_max        : 1
  boundary        : periodic
  ordering        : blocked
  warm_reps       : 2
  adapt_max_depth : 80
  replay_reps     : 2
  t_final         : 10.0
  trotter_steps   : 128
  num_times       : 135

  Command:
    /opt/anaconda3/bin/python pipelines/hardcoded/hh_staged_circuit_report.py

============================================================
```

```
HH staged circuit report, L=2

Warm: E=-0.369648170331 exact=-0.389553103297 |dE|=1.990493e-02
ADAPT: depth=79 pool=phase1_v1 stop=max_depth |dE|=1.990493e-02
Replay: E=-0.370366844427 exact=-0.389553103297 |dE|=1.918626e-02

Section semantics
- Representative view keeps high-level PauliEvolutionGate blocks.
- Expanded view performs one decomposition pass to expose term-level layers.
- Dynamics pages show one macro-step only; proxy totals summarize the full repeat count.
```

```
L=2 warm HH-HVA

Circuit summary
  - num_qubits: 6
  - depth: 17
  - size: 18
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 16, "x": 2}

Metadata
  - ansatz: hh_hva_ptw
  - reps: 2
  - energy: -0.369648170331
  - exact_energy: -0.389553103297
  - delta_abs: 0.0199049329664

Notes
  - Representative view keeps PauliEvolutionGate blocks intact.
  - Expanded view applies one circuit-definition decomposition pass.
```

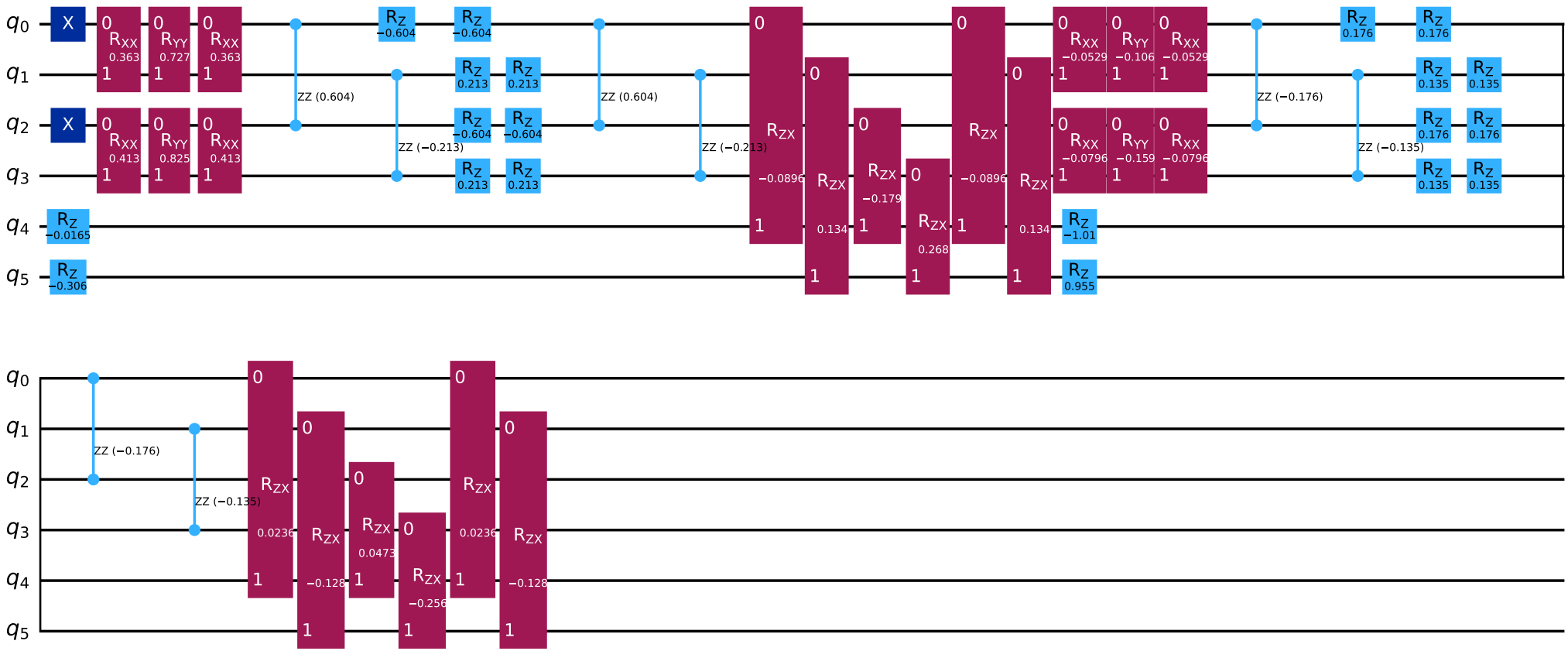
L=2 warm HH-HVA

Representative view

```
Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.
```

L=2 warm HH-HVA

Expanded view

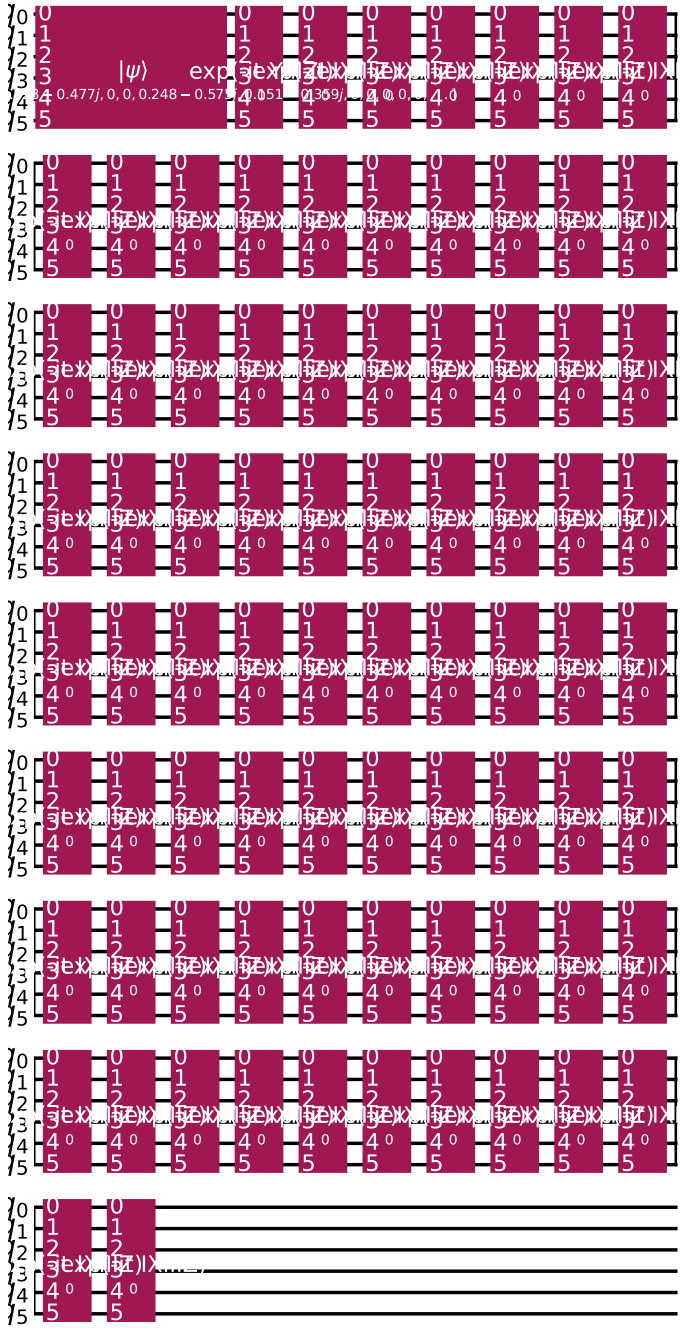


Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.

```
L=2 ADAPT-VQE

Circuit summary
  - num_qubits: 6
  - depth: 80
  - size: 80
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 79, "initialize": 1}

Metadata
  - depth: 79
  - pool_type: phase1_v1
  - continuation_mode: phase1_v1
  - energy: -0.369648170331
  - exact_energy: -0.389553103297
  - stop_reason: max_depth

Notes
  - Circuit uses the actual selected ADAPT generators and optimized theta values.
  - Reference state is the warm-stage output handed to ADAPT.
```

L=2 ADAPT-VQE

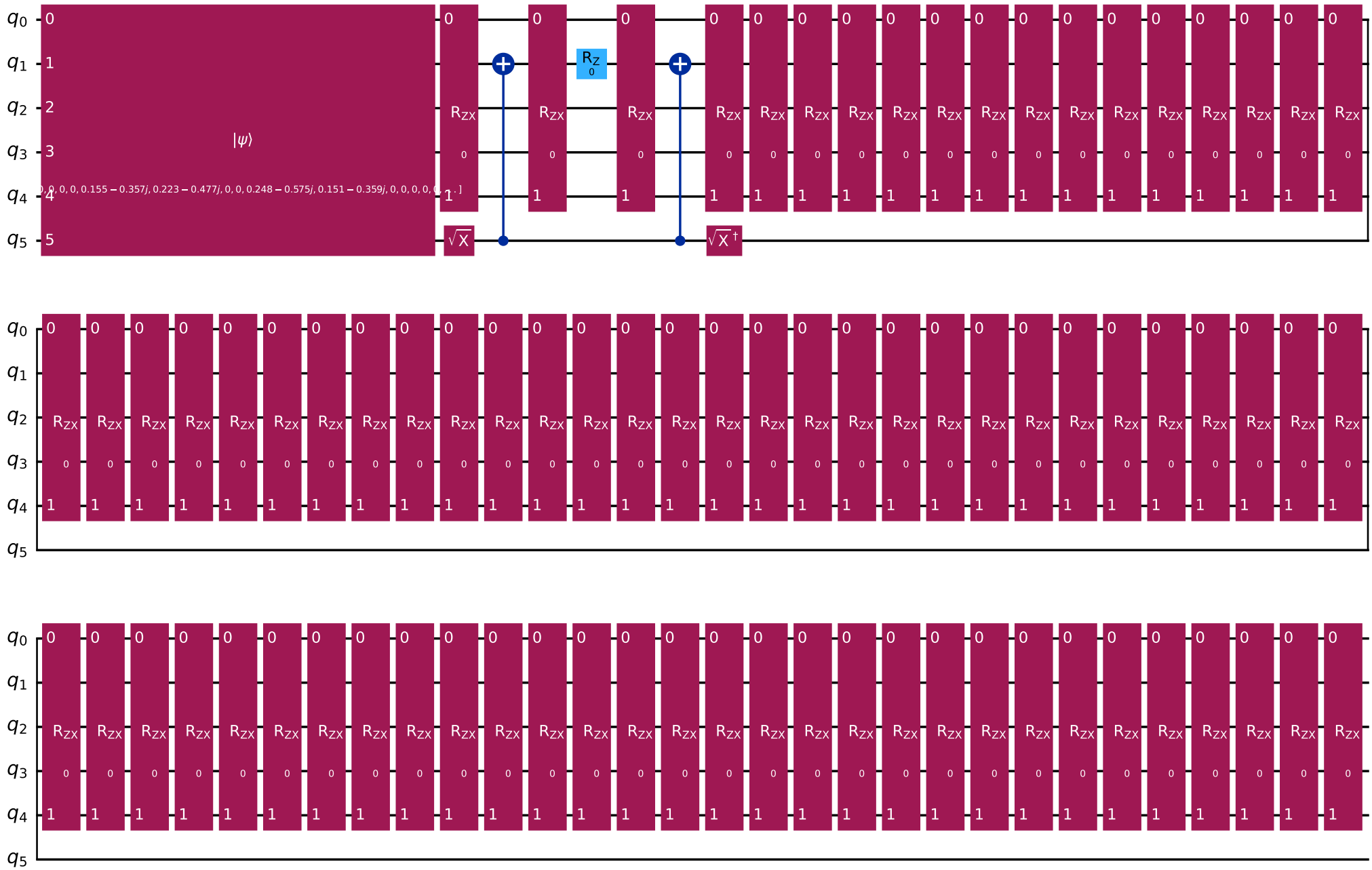
Representative view



Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

L=2 ADAPT-VQE

Expanded view

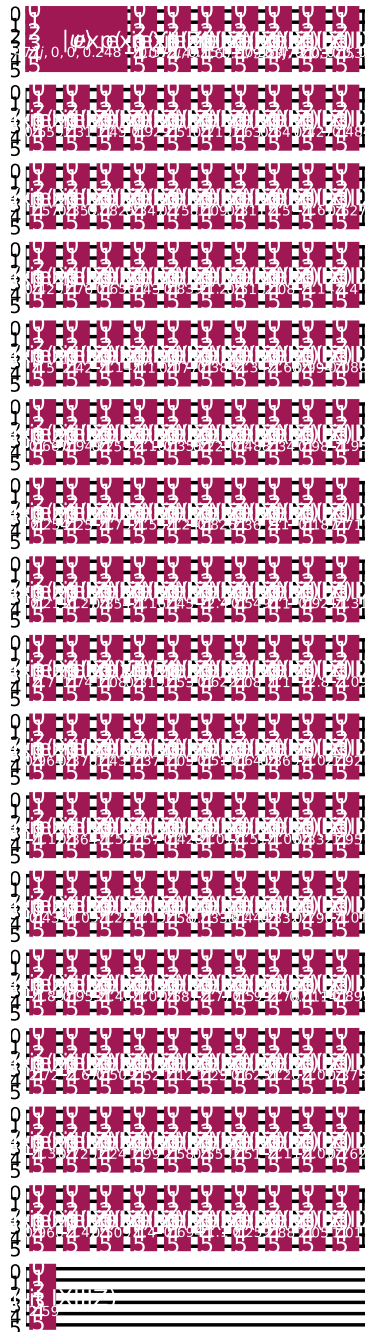


Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

```
L=2 matched-family replay

Circuit summary
  - num_qubits: 6
  - depth: 159
  - size: 159
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 158, "initialize": 1}

Metadata
  - family_info: {"fallback_family": "full_meta", "fallback_used": true, "requested": "match_adapt",
    "resolution_source": "fallback_family", "resolved": "full_meta", "warning": "Could not resolve family from
    input metadata; using fallback family 'full_meta'."}
  - reps: 2
  - seed_policy: residual_only
  - energy: -0.370366844427
  - exact_energy: -0.389553103297
  - stop_reason: converged

Notes
  - Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
  - This is the conventional non-ADAPT VQE stage in the staged HH workflow.
```

L=2 matched-family replay

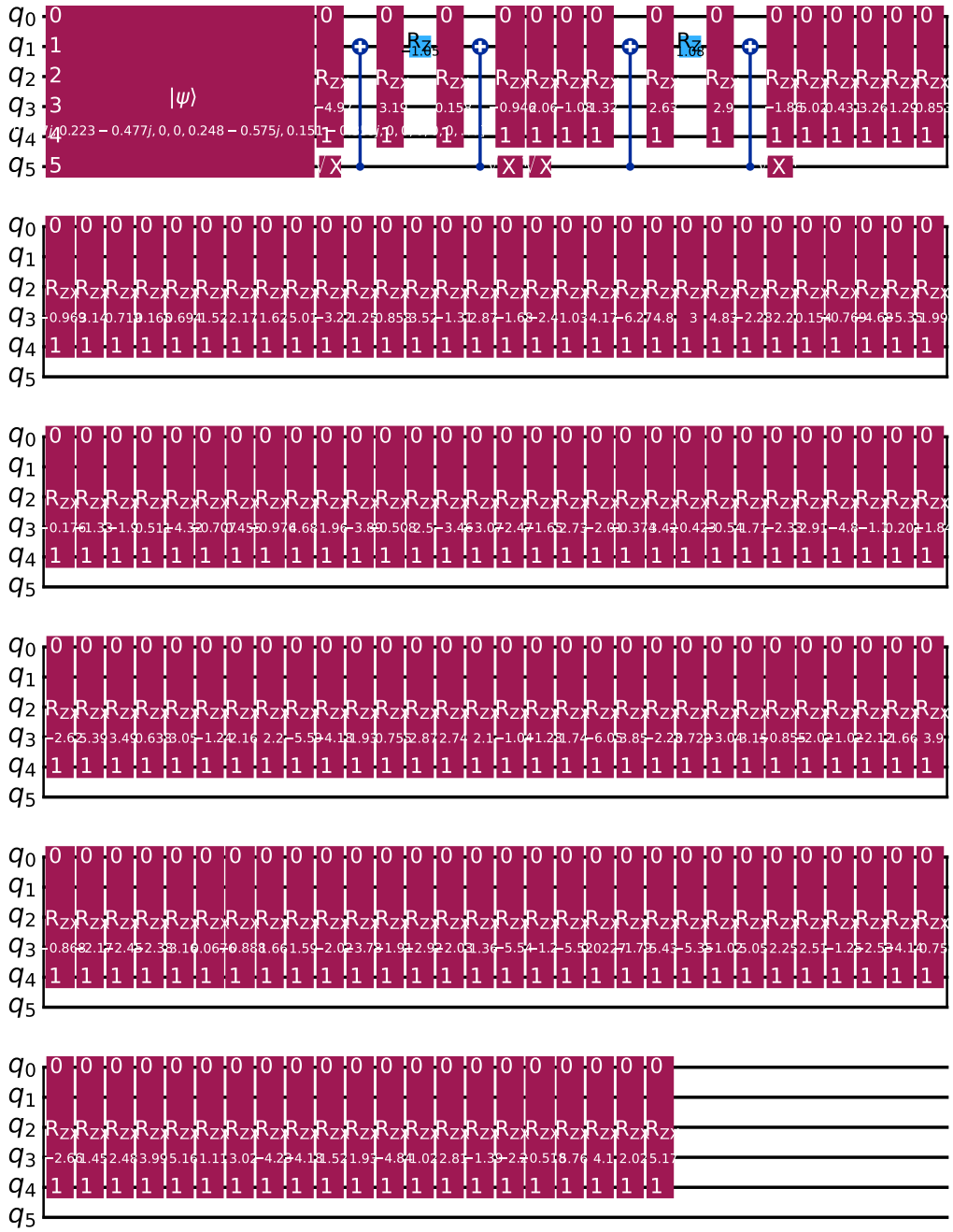
Representative view



Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
This is the conventional non-ADAPT VQE stage in the staged HH workflow.

L=2 matched-family replay

Expanded view

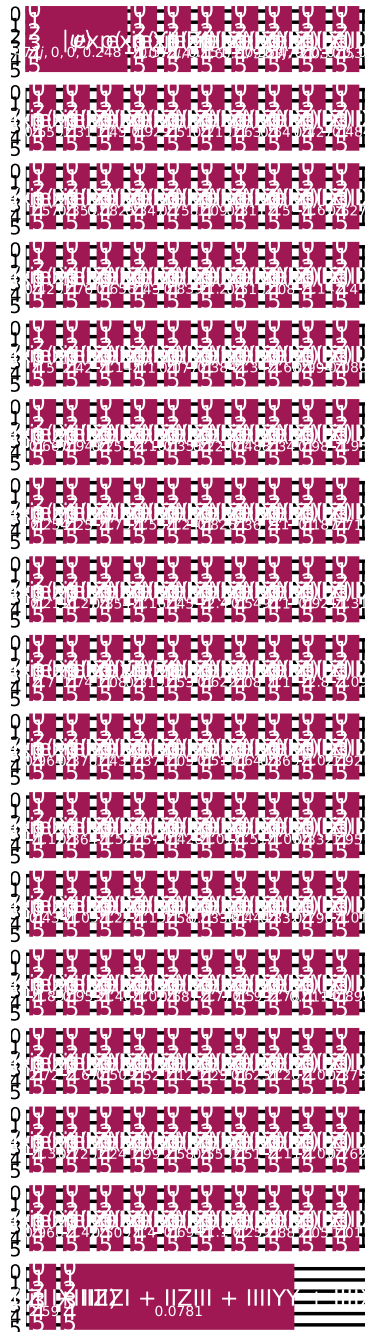


Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
This is the conventional non-ADAPT VQE stage in the staged HH workflow.

```
L=2 SUZUKI2 dynamics macro-step

Circuit summary
  - num_qubits: 6
  - depth: 160
  - size: 160
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 159, "initialize": 1}

Metadata
  - repeat_count: 128
  - macro_step_time: 0.078125
  - t_final: 10
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 40, "sq_proxy_total": 82, "term_exp_count_total": 34}
  - proxy_total: {"cx_proxy_total": 5120, "sq_proxy_total": 10496, "term_exp_count_total": 4352}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=2 SUZUKI2 dynamics macro-step

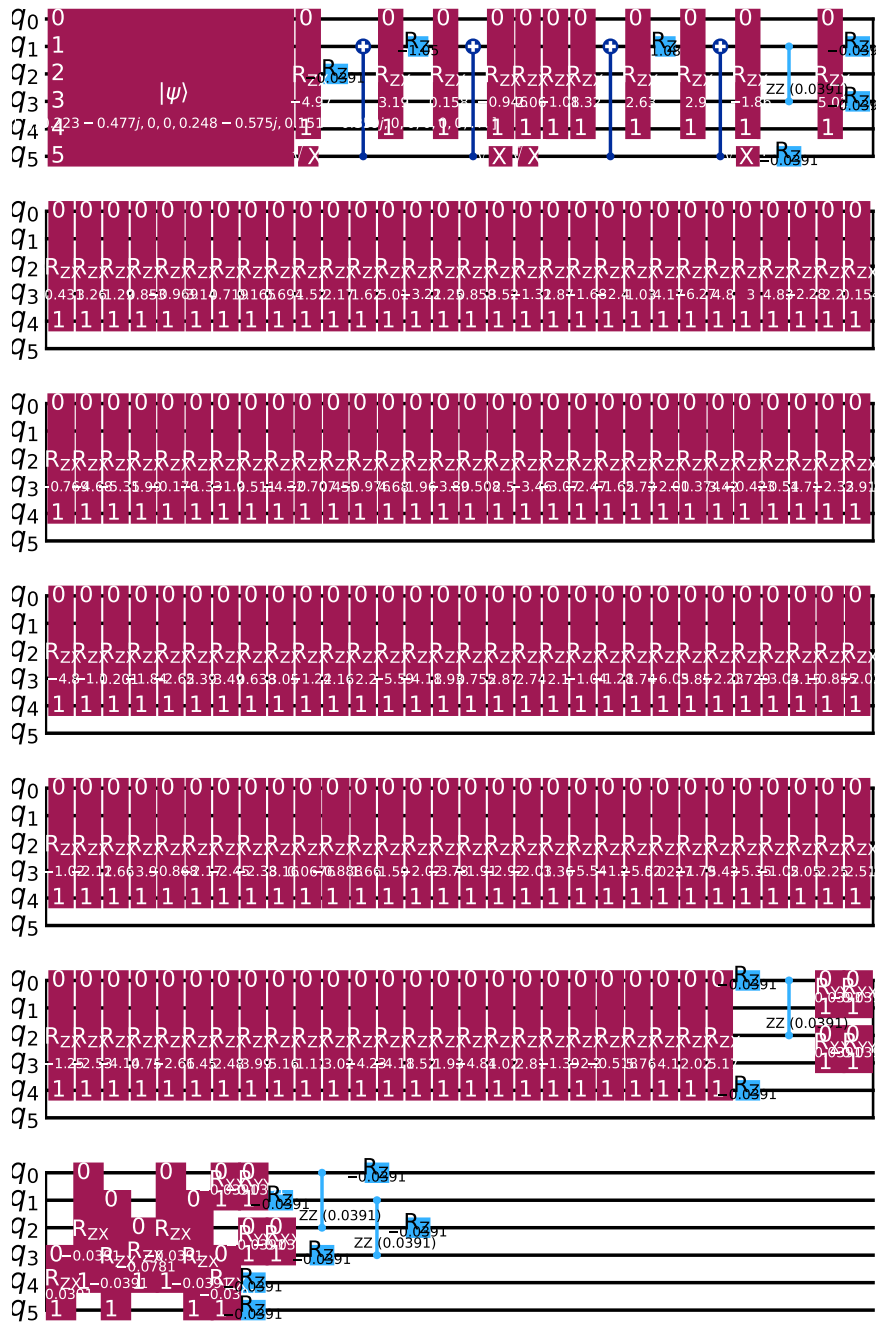
Representative macro-step



Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

L=2 SUZUKI2 dynamics macro-step

Expanded macro-step

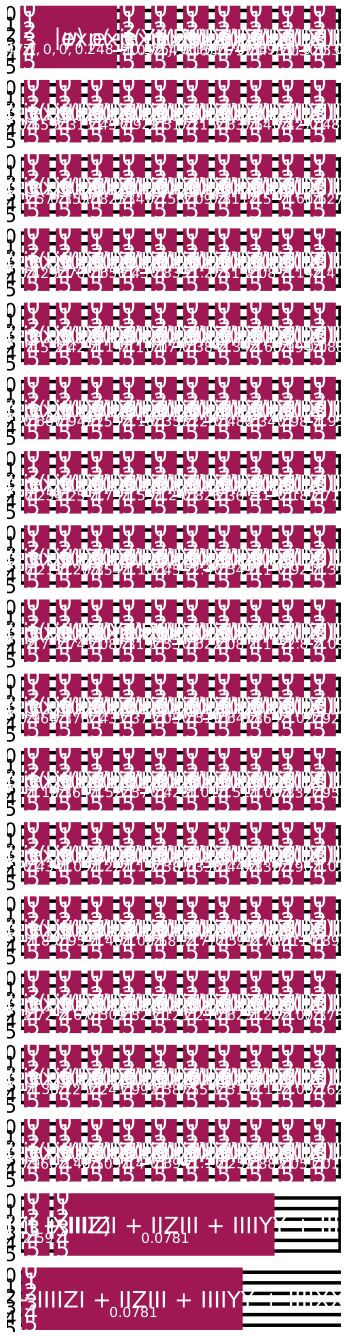


Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

```
L=2 CFQM4 dynamics macro-step

Circuit summary
  - num_qubits: 6
  - depth: 161
  - size: 161
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 160, "initialize": 1}

Metadata
  - repeat_count: 128
  - macro_step_time: 0.078125
  - t_final: 10
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 80, "sq_proxy_total": 164, "term_exp_count_total": 68}
  - proxy_total: {"cx_proxy_total": 10240, "sq_proxy_total": 20992, "term_exp_count_total": 8704}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=2 CFQM4 dynamics macro-step

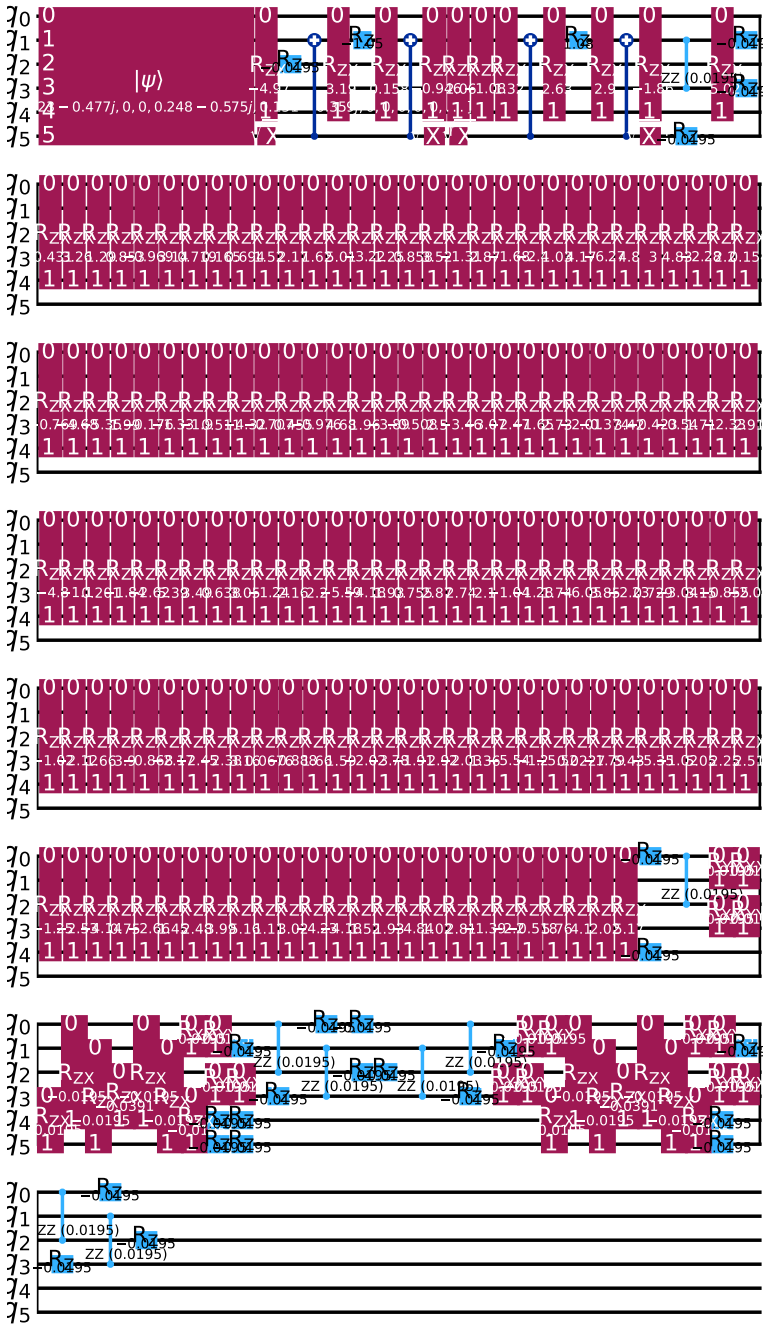
Representative macro-step



Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

L=2 CFQM4 dynamics macro-step

Expanded macro-step



Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

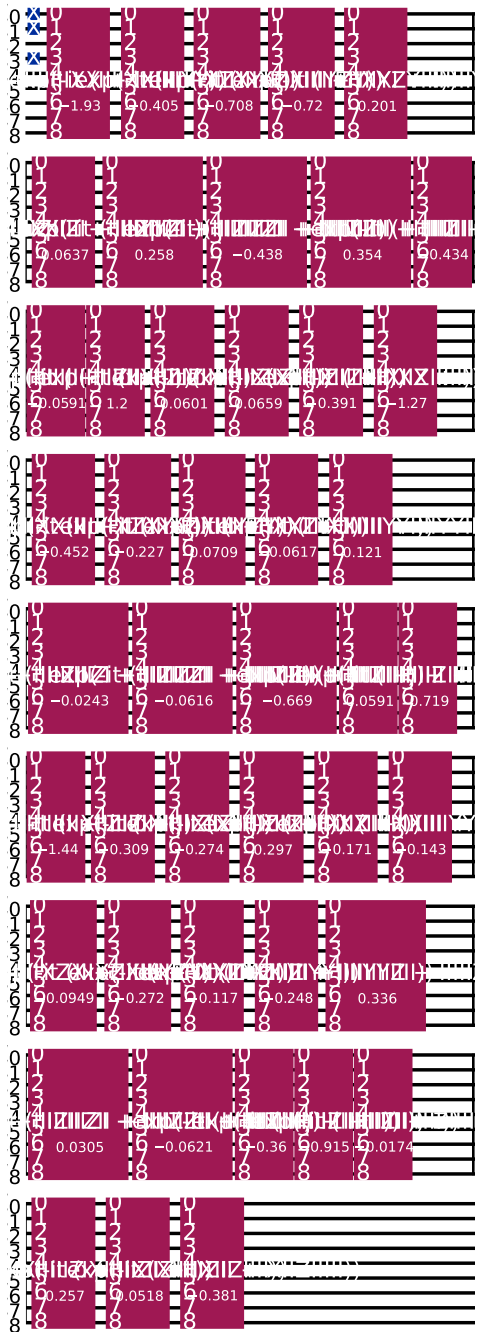
```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein (HH)
  Ansatz          : warm: hh_hva_ptw; ADAPT: phase1_v1; final: matched-family replay
  Drive enabled   : False

  t (hopping)     : 1.0
  U (interaction) : 2.0
  dv (disorder)   : 0.0

  L               : 3
  omega0          : 1.0
  g_ep            : 1.0
  n_ph_max        : 1
  boundary        : periodic
  ordering        : blocked
  warm_reps       : 3
  adapt_max_depth : 120
  replay_reps     : 3
  t_final         : 15.0
  trotter_steps   : 192
  num_times       : 201

  Command:
    /opt/anaconda3/bin/python pipelines/hardcoded/hh_staged_circuit_report.py

============================================================
```

```
HH staged circuit report, L=3

Warm: E=-0.580719361358 exact=-0.6988083851 |dE|=1.180890e-01
ADAPT: depth=120 pool=phase1_v1 stop=max_depth |dE|=1.059018e-01
Replay: E=-0.597671676655 exact=-0.6988083851 |dE|=1.011367e-01

Section semantics
- Representative view keeps high-level PauliEvolutionGate blocks.
- Expanded view performs one decomposition pass to expose term-level layers.
- Dynamics pages show one macro-step only; proxy totals summarize the full repeat count.
```

```
L=3 warm HH-HVA

Circuit summary
  - num_qubits: 9
  - depth: 46
  - size: 48
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 45, "x": 3}

Metadata
  - ansatz: hh_hva_ptw
  - reps: 3
  - energy: -0.580719361358
  - exact_energy: -0.6988083851
  - delta_abs: 0.118089023741

Notes
  - Representative view keeps PauliEvolutionGate blocks intact.
  - Expanded view applies one circuit-definition decomposition pass.
```

L=3 warm HH-HVA

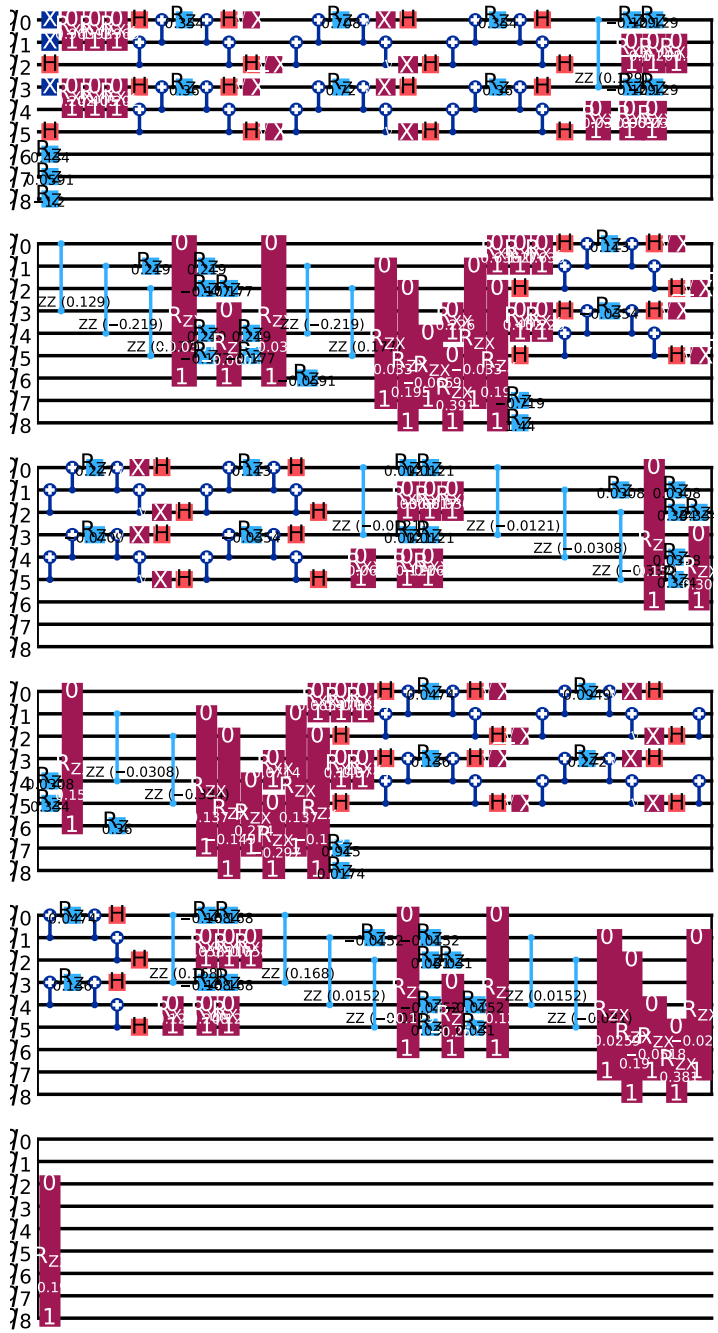
Representative view



Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.

L=3 warm HH-HVA

Expanded view

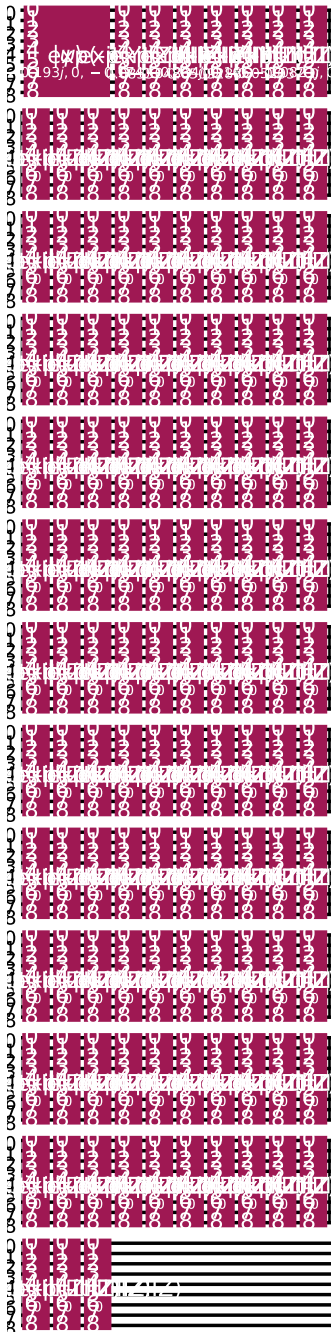


Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.

```
L=3 ADAPT-VQE

Circuit summary
  - num_qubits: 9
  - depth: 121
  - size: 121
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 120, "initialize": 1}

Metadata
  - depth: 120
  - pool_type: phase1_v1
  - continuation_mode: phase1_v1
  - energy: -0.592906622723
  - exact_energy: -0.6988083851
  - stop_reason: max_depth

Notes
  - Circuit uses the actual selected ADAPT generators and optimized theta values.
  - Reference state is the warm-stage output handed to ADAPT.
```

L=3 ADAPT-VQE

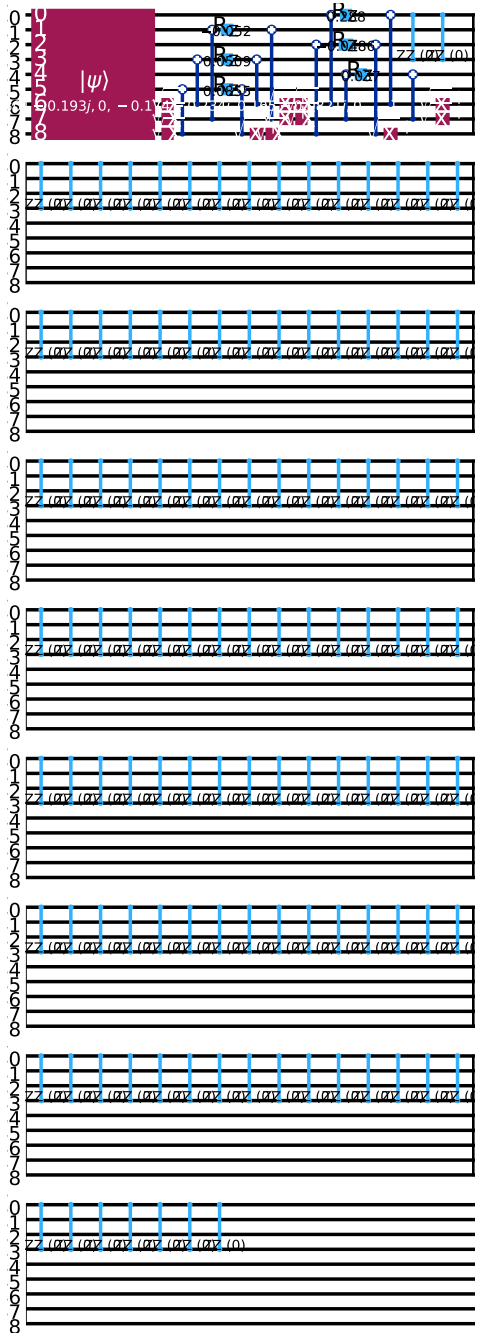
Representative view



Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

L=3 ADAPT-VQE

Expanded view



Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

```
L=3 matched-family replay

Circuit summary
  - num_qubits: 9
  - depth: 361
  - size: 361
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 360, "initialize": 1}

Metadata
  - family_info: {"fallback_family": "full_meta", "fallback_used": true, "requested": "match_adapt",
    "resolution_source": "fallback_family", "resolved": "full_meta", "warning": "Could not resolve family from
    input metadata; using fallback family 'full_meta'."}
  - reps: 3
  - seed_policy: residual_only
  - energy: -0.597671676655
  - exact_energy: -0.6988083851
  - stop_reason: converged

Notes
  - Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
  - This is the conventional non-ADAPT VQE stage in the staged HH workflow.
```

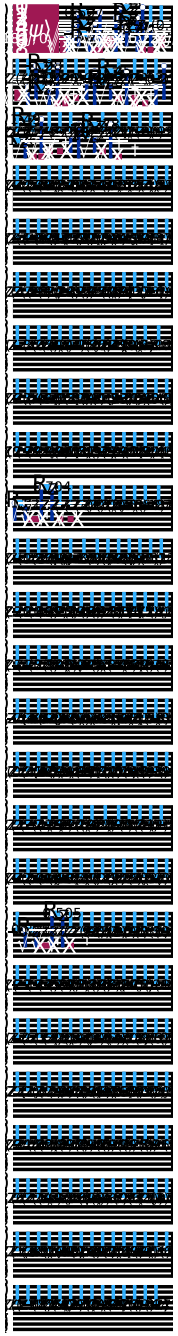
L=3 matched-family replay

Representative view

|0⟩
|+⟩
|+⟩
|+

L=3 matched-family replay

Expanded view



Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
This is the conventional non-ADAPT VQE stage in the staged HH workflow.

```
L=3 SUZUKI2 dynamics macro-step

Circuit summary
  - num_qubits: 9
  - depth: 362
  - size: 362
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 361, "initialize": 1}

Metadata
  - repeat_count: 192
  - macro_step_time: 0.078125
  - t_final: 15
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 100, "sq_proxy_total": 182, "term_exp_count_total": 62}
  - proxy_total: {"cx_proxy_total": 19200, "sq_proxy_total": 34944, "term_exp_count_total": 11904}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

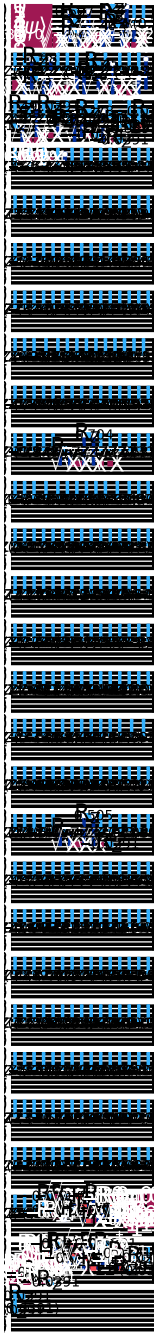
L=3 SUZUKI2 dynamics macro-step

Representative macro-step

Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

L=3 SUZUKI2 dynamics macro-step

Expanded macro-step



Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

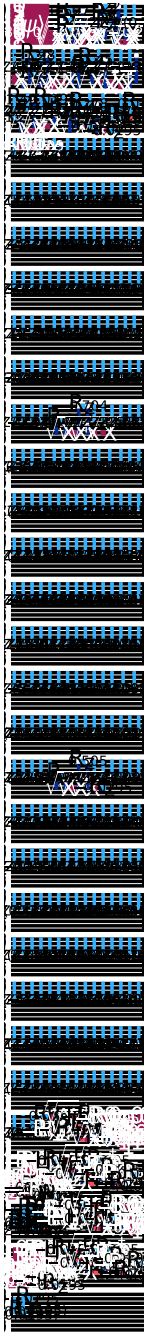
```
L=3 CFQM4 dynamics macro-step

Circuit summary
  - num_qubits: 9
  - depth: 363
  - size: 363
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 362, "initialize": 1}

Metadata
  - repeat_count: 192
  - macro_step_time: 0.078125
  - t_final: 15
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 200, "sq_proxy_total": 364, "term_exp_count_total": 124}
  - proxy_total: {"cx_proxy_total": 38400, "sq_proxy_total": 69888, "term_exp_count_total": 23808}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=3 CFQM4 dynamics macro-step

Representative macro-step

L=3 CFQM4 dynamics macro-step

Expanded macro-step



Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.